Smart toys pipeline

Smart toys pipeline

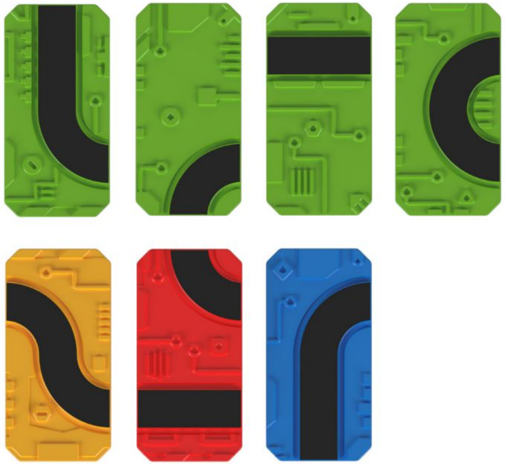
1. Image rectification

Smart toys pipeline

2. Tiles classification

Smart toys pipeline

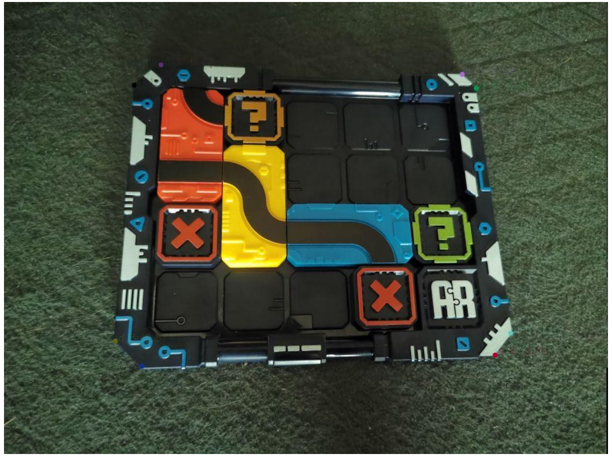
3. Consistency check: utilize prior information about the board

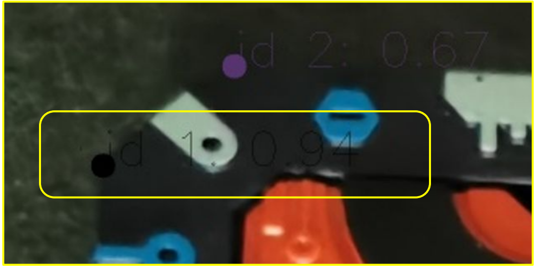
4. Make a decision: reject prediction or not

Image rectification pipeline

1. Detect 8 corners



2. Find 4 principal points for homography matrix calculation: 1 with highest score from 4 pairs



3. Apply homography to image

Image rectification pipeline

NOTE: If both 2 points of a board corner are invisible, then we won't rectify, because of high possibility to get an error in homography. Predicted points with scores >= *vis_threshold* are assumed to be visible.

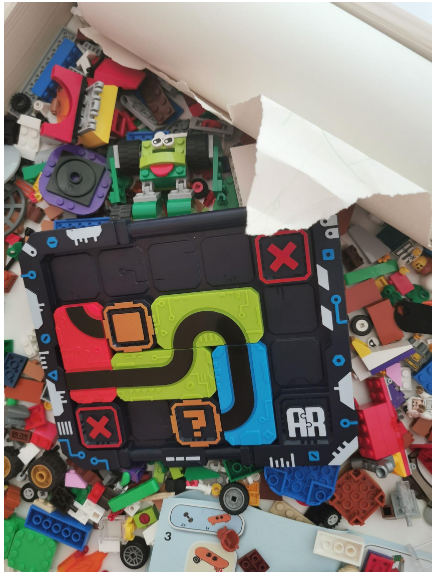


Image rectification quality metric

Accuracy = |{Well rectified images}| / |{Images with all visible corners}|

Image is supposed to be well rectified if:

- All predicted principal points have scores more or equal to *vis_threshold* and belong to the *distance_threshold*-neighborhood of annotated points.
- We calculate distances between annotated and predicted points in real world coordinates on board plane.
- We choose *distance_threshold = 2.25 mm*.

In the repository *smartnv-corners-train* this metric is called *2.25mm max principal distance accuracy.*

We also calculate the same metric with other thresholds: 1, 1.5.

Image rectification quality metric

Additional metrics, that we use:

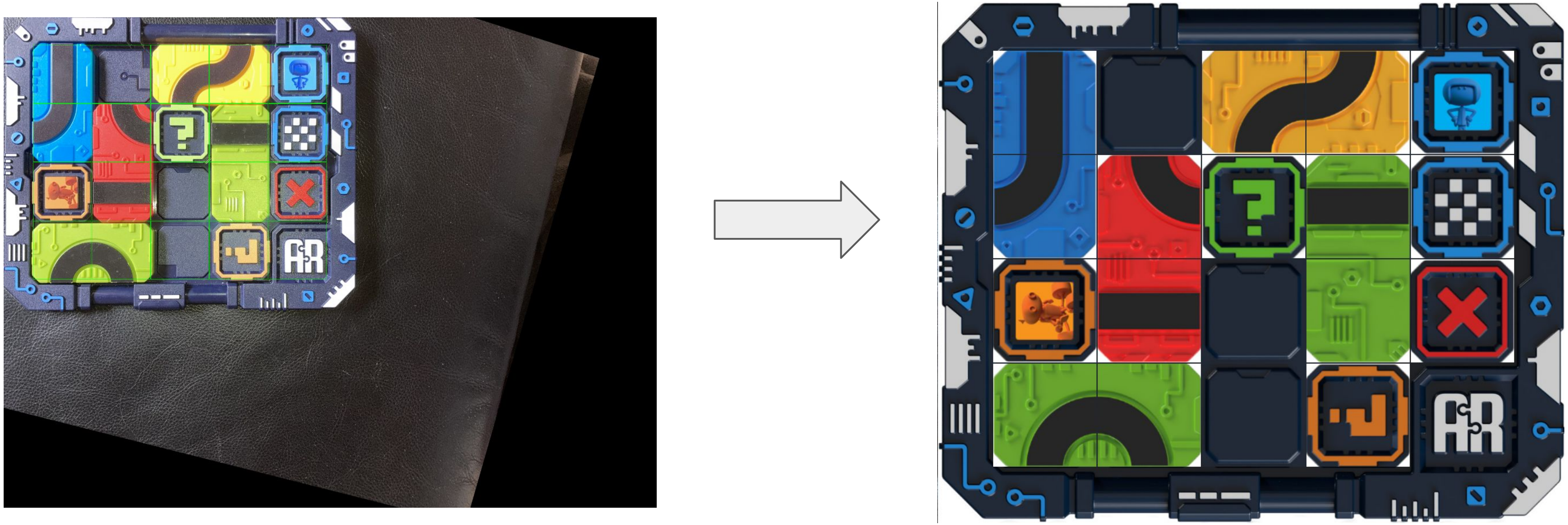
- *10-pixel accuracy* = |{predicted points in 10-pixel-neighbourhood of target ones}| / |{all visible points}|
- *Average distance* - average distance between predicted points and target ones in real world coordinates on board plane (mm).
- *Average 1.5mm distance accuracy* = |{predicted points in 1.5-mm-neighbourhood of target ones}| / |{all visible points}|
- *Visibility accuracy* = accuracy of binary classification visible/non-visible (predicted point is visible if its score is more or equal to *vis_threshold*)
- *Class-wise accuracy* - *10-pixel accuracy* per class
- *Class-wise distance* - *Average distance* per class
- *Class-wise 1.5mm distance accuracy* - *Average 1.5mm distance accuracy* per class

Tiles classification pipeline

A batch of 19 tiles images is fed to the classifier that determines types and particular rotations of the given tiles as well as provides confidence scores.

Tiles classification pipeline

In more technical words, for each image the classifier model predicts probability distribution among classes. By default, the predicted class is that with the highest probability (score).



Tiles classification quality metric

Accuracy = |{Correctly classified images}| / |{All images}|

Image is supposed to be correctly classified if:

- Predicted class label with the highest score is equal to target label.

In the repository *smartnv-classifier-train* this metric is called *Acc@1.*

We also calculate *Acc@5* metric where requirements for prediction os weaker: image is supposed to be correctly classified if one of the 5 predicted class labels with the highest scores is equal to target label.

Consistency check: utilize prior information about the board

We can check whether final predictions is correct or even fix our predictions relying on prior information about the board. Firstly, we know the max number of tiles of each class that can appear on the board. Secondly, we know that some of square tiles are belong to doubled real tiles.

Make a decision: reject the prediction or not

We reject our final prediction and ask a user to make another photo, if:

1. Min(scores of 4 principal corners) < corner_score_threshold

   or

2. Min (scores of 19 tiles’ classes predictions) < tiles_score_threshold

   or

3. Consistency check failed

End-to-end quality metrics

Precision = |{Correct predictions}| / |{All predictions}|

Recall = |{Correct predictions}| / |{All images}|

Final prediction is an row-major array of board tiles classes (of length 19).

Prediction is supposed to be correct if all tiles are classified correctly. Remember, that a number of final predictions depends on rejection rule, described early.

Varying corners and tiles thresholds, we get different Precision, Recall values. So we calculate this metrics for values of thresholds from 2-dimensional uniform grid. We can choose thresholds related to the preferable Precision, Recall values.

End-to-end quality metrics

We print calculated values of Precision and Recall as well as related thresholds:

Precision	Recall	Corners score threshold	Tiles score threshold
1.0000	0.9423	0.2000	0.4500
1.0000	0.9423	0.2000	0.5000
1.0000	0.9423	0.2500	0.4500
1.0000	0.9423	0.2500	0.5000
1.0000	0.9423	0.3000	0.4500
1.0000	0.9423	0.3000	0.5000
0.9960	0.9577	0.2000	0.2500
0.9960	0.9577	0.2500	0.2500
0.9960	0.9577	0.3000	0.2500
0.9922	0.9731	0.2000	0.0000
0.9922	0.9731	0.2000	0.0500
0.9922	0.9731	0.2500	0.0000
0.9922	0.9731	0.2500	0.0500
0.9922	0.9731	0.3000	0.0000
0.9922	0.9731	0.3000	0.0500

End-to-end quality metrics

We also draw calculated values of Precision and Recall: